Alternative proof of Lemma 3

Recall that

$$T_1(x) = x_1 + x_2, \qquad \Pi = \mathrm{cone}\{(1,0),(1,-1)\}, \qquad \widehat{A} = \mathrm{cl}(A - \Pi),$$

and define

$$\mathcal{F} = \{a \in \mathbf{R}^2 : 0 \le a_2 \le a_1\}.$$

Lemma 3 (Calibration of the competitor)**.** *Suppose that $A \subseteq \mathbf{R}^2$ is closed, convex, and coordinatewise lower, $P_0(A) = 1 - \alpha$, and A has a finite-support direction a satisfying $a_2 > a_1$. Then there exists a finite number b such that*

$$P_0\big(\widehat{A} \cap \{x : T_1(x) \le b\}\big) = 1 - \alpha.$$

For any such b, define

$$A' = \widehat{A} \cap \{x : T_1(x) \le b\}.$$

Then A' is closed, convex, and coordinatewise lower, satisfies $P_0(A') = 1 - \alpha$, and every nonzero finite-support direction of A' lies in $\mathcal{F}$. Consequently, the test $\phi_{A'}(x) = I\{x \notin A'\}$ has level α, attained at $\mu = 0$.

Proof. By Lemma 2,

$$P_0(\widehat{A}) > P_0(A) = 1 - \alpha.$$

The map

$$t \longmapsto P_0\big(\widehat{A} \cap \{x : T_1(x) \le t\}\big)$$

is continuous because $P_0\{T_1(X) = t\} = 0$ for every t. It converges to zero as $t \to -\infty$ and to $P_0(\widehat{A})$ as $t \to \infty$. Hence there exists a finite b for which the displayed probability equals $1 - \alpha$.

The set A' is closed, convex, and coordinatewise lower because it is the intersection of two sets with these properties. Its null probability equals $1 - \alpha$ by construction.

It remains to characterize its finite-support directions. We first show that A' is Π-lower. Let $x \in A'$ and $p \in \Pi$. Lemma 2 implies that $x - p \in \widehat{A}$. Moreover, $T_1(p) = p_1 + p_2 \ge 0$ for every $p \in \Pi$, and therefore

$$T_1(x - p) = T_1(x) - T_1(p) \le T_1(x) \le b.$$

Thus $x - p \in A'$.

Now let $a \neq 0$ satisfy $h_{A'}(a) < \infty$, and fix $x \in A'$. Since A' is coordinatewise lower, $x - te_j \in A'$ for every $t \ge 0$ and $j = 1, 2$. If $a_j < 0$, then

$$a'(x - te_j) = a'x - ta_j \longrightarrow +\infty,$$

contradicting $h_{A'}(a) < \infty$. Hence $a_1 \ge 0$ and $a_2 \ge 0$.

Because $(1,-1) \in \Pi$ and A' is Π-lower,

$$x - t(1,-1) \in A' \qquad \text{for every } t \ge 0.$$

If $a_2 > a_1$, then

$$a'\{x - t(1,-1)\} = a'x + t(a_2 - a_1) \longrightarrow +\infty,$$

again contradicting finite support. Therefore $a_2 \le a_1$. We have shown that every nonzero a with $h_{A'}(a) < \infty$ satisfies

$$0 \le a_2 \le a_1,$$

so $a \in \mathcal{F}$.

Finally, the complement of A' is coordinatewise upper. For every $\mu \in \Lambda_0$, we have $\mu \le 0$. Coupling $X = \mu + Z$ with $Z \sim N_2(0, I_2)$ gives

$$P_\mu(X \notin A') \le P_0(Z \notin A') = \alpha.$$

Equality holds at the finite null point $\mu = 0$, which proves the last claim. □